

Video 5.1.1

Operation #3 of 3: Matrix multiplication.

First key point:

You can only multiply matrices when their shapes “match”.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

* no. of column of left matrix = no. of rows of left matrix.

Operation #3 of 3: Matrix multiplication.

First points:

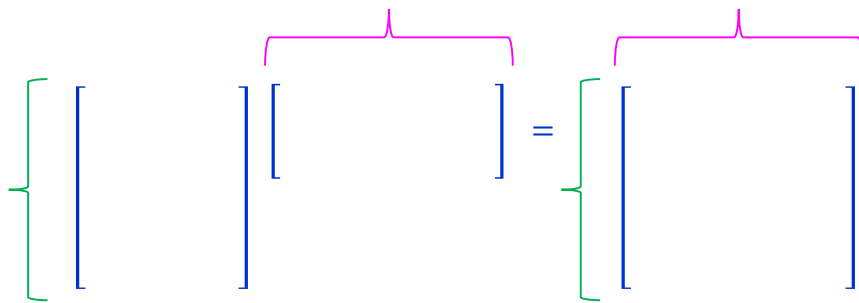
You can only multiply matrices when their shapes “match”.

$$\left[\begin{array}{c} \vdots \\ \vdots \end{array} \right]$$

For the product of these two matrices to make sense (we say “well-defined”) we need the number of columns of the left matrix ...

]

... to equal the number of rows of the right matrix...



... and then:

- the **number of rows of the left matrix** becomes the number of rows of the resulting product
- and the **number of columns of the right matrix** becomes the number of columns of the resulting product.

$$“(m \times \cancel{n}) \cdot (\cancel{n} \times p) = (m \times p)”$$

So let's go through how you multiply now. For the discussion we will consider two matrices:

$$A = [a_{ij}]_{m \times p} \quad \text{and} \quad B = [b_{ij}]_{p \times n}.$$

Note that:

$$(\# \text{ of columns of } A) = (\# \text{ of rows of } B)$$

so the product can be defined.

We will introduce this by saying how you separately construct each entry of the resulting product.

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ a_{41} & a_{42} & a_{43} \\ a_{51} & a_{52} & a_{53} \\ a_{61} & a_{62} & a_{63} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & b_{13} & b_{14} \\ b_{21} & b_{22} & b_{23} & b_{24} \\ b_{31} & b_{32} & b_{33} & b_{34} \end{bmatrix} = \begin{bmatrix} ? \\ ? \\ ? \\ ? \\ ? \\ ? \end{bmatrix}$$

The rule for matrix multiplication

To construct the (i, j) -entry of the product matrix $\mathbf{AB}$

1. Take the i -th row of the left matrix $\mathbf{A}$

$$a_{i1} \ a_{i2} \ \dots \ a_{ip}$$

2. Take the j -th column of the right matrix $\mathbf{B}$

$$b_{1j} \ b_{2j} \ \dots \ b_{pj}$$

3. Combine them by going along these lists multiplying corresponding terms then adding up:

$$a_{i1}b_{1j} + a_{i2}b_{2j} + \dots + a_{ip}b_{pj}$$

Quick check of understanding:

Consider the following case with $m = 6, p = 3, n = 4$.

What is the entry in the $(5, 2)$ position of the product?

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ a_{41} & a_{42} & a_{43} \\ a_{51} & a_{52} & a_{53} \\ a_{61} & a_{62} & a_{63} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & b_{13} & b_{14} \\ b_{21} & b_{22} & b_{23} & b_{24} \\ b_{31} & b_{32} & b_{33} & b_{34} \end{bmatrix} = \begin{bmatrix} & & & \\ & & & \\ & & & \\ & & & \\ & ? & & \\ & & & \end{bmatrix}$$

1. You take the 5-th row of the left matrix.
2. Then you take the 2-nd column of the right matrix.
3. Then combine: $a_{51}b_{12} + a_{52}b_{22} + a_{53}b_{32}$.

Quick check of understanding:

Consider the following case with $m = 6, p = 3, n = 4$.

What is the entry in the (1,3) position of the product?

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ a_{41} & a_{42} & a_{43} \\ a_{51} & a_{52} & a_{53} \\ a_{61} & a_{62} & a_{63} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & b_{13} & b_{14} \\ b_{21} & b_{22} & b_{23} & b_{24} \\ b_{31} & b_{32} & b_{33} & b_{34} \end{bmatrix} = \begin{bmatrix} ? \\ \\ \\ \\ \\ \end{bmatrix}$$

1. You take the 1-st row of the left matrix.
2. Then you take the 3-rd column of the right matrix.
3. Then combine: $a_{11}b_{13} + a_{12}b_{23} + a_{13}b_{33}$.

Quick check of understanding:

Consider the following case with $m = 6, p = 3, n = 4$.

What is the entry in the (4,4) position of the product?

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ a_{41} & a_{42} & a_{43} \\ a_{51} & a_{52} & a_{53} \\ a_{61} & a_{62} & a_{63} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & b_{13} & b_{14} \\ b_{21} & b_{22} & b_{23} & b_{24} \\ b_{31} & b_{32} & b_{33} & b_{34} \end{bmatrix} = \begin{bmatrix} \\ \\ \\ ? \\ \\ \end{bmatrix}$$

1. You take the 4-th row of the left matrix.
2. Then you take the 4-th column of the right matrix.
3. Then combine: $a_{41}b_{14} + a_{42}b_{24} + a_{43}b_{34}$.

OK, let's write down a formal definition.

Operation #3 of 3: Matrix multiplication.

Definition (Multiplication of two matrices).

Take two matrices $A \in \mathbb{M}_{m \times p}(\mathbb{R})$ and $B \in \mathbb{M}_{p \times n}(\mathbb{R})$.

Assume

$$A = [a_{ij}]_{m \times p} \text{ and } B = [b_{ij}]_{p \times n}.$$

The product of these two matrices is the $m \times n$ matrix

$$AB \in \mathbb{M}_{m \times n}(\mathbb{R})$$

whose (i, j) -entry is given by the formula

$$a_{i1}b_{1j} + \cdots + a_{ip}b_{pj} = \sum_{k=1}^p a_{ik}b_{kj}.$$

Take note there is a compact way to write the entry using summation notation. This will be very useful!

Example

$$\begin{array}{c}
 \begin{array}{cc} 2 \times 3 & 3 \times 2 \Rightarrow \text{can multiply!} \end{array} \\
 \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & 3 \\ -1 & -2 \end{pmatrix} \\
 = \begin{pmatrix} 1 \times 1 + 2 \times 2 + 3 \times (-1) & 1 \times 1 + 2 \times 3 + 3 \times (-2) \\ 4 \times 1 + 5 \times 2 + 6 \times (-1) & 4 \times 1 + 5 \times 3 + 6 \times (-2) \end{pmatrix} \\
 = \begin{pmatrix} 2 & 1 \\ 8 & 7 \end{pmatrix}
 \end{array}$$

Exercise

 3×2 2×3 $\Rightarrow$ can be multiplied

$$\begin{pmatrix} 1 & 1 \\ 2 & 3 \\ -1 & -2 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \times 1 + 1 \times 4 & 1 \times 2 + 1 \times 5 & 1 \times 3 + 1 \times 6 \\ 2 \times 1 + 3 \times 4 & 2 \times 2 + 3 \times 5 & 2 \times 3 + 3 \times 6 \\ (-1) \times 1 + (-2) \times 4 & (-1) \times 2 + (-2) \times 5 & (-1) \times 3 + (-2) \times 6 \end{pmatrix}$$

$$= \begin{pmatrix} 5 & 7 & 9 \\ 14 & 19 & 24 \\ -9 & -12 & -15 \end{pmatrix}$$

Video 5.1.2

Properties of Matrix multiplication.

We just defined a new product.

To organize our thinking about the properties of this new multiplication by asking the following question:

Question:

Does this new multiplication we just defined for matrices behave like the familiar multiplication for numbers?

Answer:

Sometimes it does.
Sometimes it doesn't.

Properties where matrix multiplication behaves like the familiar multiplication in $\mathbb{R}$.

1. There is a special matrix behaving like the number $1 \in \mathbb{R}$.
(The “identity matrix” I_n .)
2. There is a special matrix behaving like the number $0 \in \mathbb{R}$.
(The “zero matrix” $0_{p \times q}$.)
3. Multiplication is associative.
 $(AB)C = A(BC)$.
4. Multiplication is distributive.
 $(A + B)C = AC + BC$
 $A(B + C) = AB + AC$

Properties where it doesn't.

1. It isn't always defined. If the shapes don't match can't do it.
2. Multiplication is not necessarily commutative:
 $AB \neq BA$.
3. Matrices don't necessarily have inverses. (Given M there may not be a M^{-1} where $MM^{-1} = M^{-1}M = I_n$.)
4. You may have divisors of zero. Thus the equation
 $AB = 0_{p \times q}$
does not necessarily imply one of A or B is a zero matrix.

Video 5.1.3

Let's go through some details of these properties.

First:

we'll consider the properties where the matrix product behaves like the familiar product in $\mathbb{R}$.

The identity.

For example $I_5 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$.

The following theorem says that this matrix indeed behaves very much like the usual number $1 \in \mathbb{R}$.

Theorem

Let $M \in \mathbb{M}_{m \times n}(\mathbb{R})$. Then

$$I_m M = M I_n = M.$$

identity matrices of different sizes

Proof: This is a problem on the coming tutorial.

The zero

For example $0_{2 \times 4} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$.

Theorem

Let $M \in \mathbb{M}_{m \times n}(\mathbb{R})$.

The

$$0_{p \times m} M = 0_{p \times n}$$

and

$$M 0_{n \times p} = 0_{m \times p}.$$

Proof: Obvious that the product will have every entry zero with the given shape. Note carefully that the shape of the zero matrix may change in this equation.

Associativity

Let's prove associativity.

Proof:

Consider three matrices

- $A \in \mathbb{M}_{p \times q}(\mathbb{R})$ with $A = [a_{ij}]_{p \times q}$.
- $B \in \mathbb{M}_{q \times r}(\mathbb{R})$ with $B = [b_{ij}]_{q \times r}$.
- $C \in \mathbb{M}_{r \times s}(\mathbb{R})$ with $C = [c_{ij}]_{r \times s}$.

Our job is to prove that

$$A(BC) = (AB)C.$$

The two sides of this equation are matrices with the same shape $p \times s$.

So to prove the equation we just need to check that all the corresponding entries are the same. We can just do a direct calculation.

Proof (continued):

Let's focus on some entry, say the (i, j) -entry.

The (i, j) -entry of the LHS $A(BC)$

$$\begin{aligned} &= \sum_{k=1}^q a_{ik} (\text{The } (k, j)\text{-entry of } (BC)) \\ &= \sum_{k=1}^q a_{ik} \left(\sum_{l=1}^r b_{kl} c_{lj} \right) \\ &= \sum_{k=1}^q \sum_{l=1}^r a_{ik} b_{kl} c_{lj}. \end{aligned}$$

Proof (continued):

Now focus on the same (i, j) -entry of the RHS.

The (i, j) -entry of the RHS $(AB)C$

$$\begin{aligned} &= \sum_{l=1}^r (\text{The } (i, l)\text{-entry of } (AB)) c_{lj} \\ &= \sum_{l=1}^r \left(\sum_{k=1}^q a_{ik} b_{kl} \right) c_{lj} \\ &= \sum_{l=1}^r \sum_{k=1}^q a_{ik} b_{kl} c_{lj}. \end{aligned}$$

This is exactly the same as expression we calculated for the (i, j) -entry of the LHS.



Distributive properties

Theorem

Let $A, B \in \mathbb{M}_{m \times n}(\mathbb{R})$.

Let $C, D \in \mathbb{M}_{n \times q}(\mathbb{R})$.

Then

$$(A + B)C = AC + BC.$$

and

$$A(C + D) = AC + AD$$

Proof:

Just a mechanical exercise in the definitions. This will be on the coming tutorial.

Video 5.1.4

Next:

We'll consider the properties where the matrix product *behaves differently to* the familiar product in $\mathbb{R}$.

Non-commutativity

The matrix multiplication is *not commutative*, i.e. in general (but not always!), AB and BA are two different matrices even if both products exist.

Consider for example:

$$A = \begin{bmatrix} -1 & 0 \\ 2 & 3 \end{bmatrix} \text{ and } B = \begin{bmatrix} 1 & 2 \\ 3 & 0 \end{bmatrix}.$$

Then you can check that:

- $AB = \begin{bmatrix} -1 & -2 \\ 11 & 4 \end{bmatrix}$
- $BA = \begin{bmatrix} 3 & 6 \\ -3 & 0 \end{bmatrix}$

Inverses don't necessarily exist.

We know that in $\mathbb{R}$ every non-zero number $c \neq 0 \in \mathbb{R}$ has an inverse. That is, there is a number $c^{-1} \in \mathbb{R}$ such that

$$cc^{-1} = c^{-1}c = 1.$$

How do we generalize this concept to matrices?

Definition

Consider a square matrix $M \in \mathbb{M}_{m \times m}(\mathbb{R})$.

M is said to be **invertible** if there exists another matrix $M^{-1} \in \mathbb{M}_{m \times m}(\mathbb{R})$ such that

$$MM^{-1} = M^{-1}M = I_m.$$

If such a matrix exists then it is said to be the **inverse of M** .

Example:

The matrix $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ -1 & -1 & 0 \end{bmatrix}$ has inverse $A^{-1} = \begin{bmatrix} -1 & 0 & -2 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$.

Check:

$$AA^{-1} = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ -1 & -1 & 0 \end{bmatrix} \begin{bmatrix} -1 & 0 & -2 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3.$$

$$A^{-1}A = \begin{bmatrix} -1 & 0 & -2 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ -1 & -1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3.$$

Fact:

Some matrices have inverses, but some don't.

Example:

Consider the matrix $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$.

Note the following equation $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \quad (*)$.

Now if $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ did actually have some inverse $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ then we could left-multiply both sides of $(*)$ with it as follows:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

↳ would give identity matrix

Then simplify the two sides of this equation separately:

- $LHS = \left(\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \right) \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}$ *↳ multiply by identity matrix gives back itself. (because its an inverse).*
- $RHS = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$

This is a contradiction so no inverse to $\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$ exists. □

Problem

Just say you have square matrices $A, B, C, D, E \in \mathbb{M}_{p \times p}(\mathbb{R})$.

Assume they are all invertible with inverses

$A^{-1}, B^{-1}, C^{-1}, D^{-1}, E^{-1} \in \mathbb{M}_{p \times p}(\mathbb{R})$.

Just their product $ABCDE \in \mathbb{M}_{p \times p}(\mathbb{R})$ have an inverse?

Give a formula for it.

Solution:

Yes. The inverse is $E^{-1}D^{-1}C^{-1}B^{-1}A^{-1}$

Check:

$$\begin{aligned} & E^{-1}D^{-1}C^{-1}B^{-1}A^{-1}ABCDE \\ &= E^{-1}D^{-1}C^{-1}B^{-1}E C D E \\ &= E^{-1}D^{-1}C^{-1}C D E \\ &= E^{-1}D^{-1}D E \\ &= E^{-1}E \\ &= I_p \end{aligned}$$

Video 5.1.5

Properties that relate Matrix multiplication and scalar multiplication.

Theorem

Let $A \in \mathbb{M}_{m \times n}(\mathbb{R})$.

Let $B \in \mathbb{M}_{n \times p}(\mathbb{R})$.

Let $r \in \mathbb{R}$.

Then

$$r(AB) = (rA)B = A(rB).$$

Proof:

Straightforward tedious exercise in the definitions.

A nice trick for thinking about matrix multiplication: dividing into blocks.

Video 5.1.6

This is a trick that can help us understand directly what matrix multiplication does in certain situations.

The best way to introduce this is with an example.

Example

Consider:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{bmatrix}$$

A direct calculation gives:

$$\begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} & a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32} \\ a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} & a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32} \\ a_{31}b_{11} + a_{32}b_{21} + a_{33}b_{31} & a_{31}b_{12} + a_{32}b_{22} + a_{33}b_{32} \end{bmatrix}$$

Example (continued)

Or we can group the matrices into blocks first. There are many different ways to do this, but we'll just show one:

$$\begin{bmatrix} \boxed{A_{11}} & \boxed{A_{12}} \\ \boxed{A_{21}} & \boxed{A_{22}} \\ \boxed{A_{31}} & \boxed{A_{32}} \end{bmatrix} \begin{bmatrix} \boxed{B_{11}} & \boxed{B_{12}} \\ \boxed{B_{21}} & \boxed{B_{22}} \end{bmatrix}$$

After we group we think of the new submatrices as individual entries:

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$$

Then we multiply them as normal, only now they are matrices multiplying, not real numbers:

$$= \begin{bmatrix} A_{11}B_{11} + A_{12}B_{21} & A_{11}B_{12} + A_{12}B_{22} \\ A_{21}B_{11} + A_{22}B_{21} & A_{21}B_{12} + A_{22}B_{22} \end{bmatrix}$$

Example (continued)

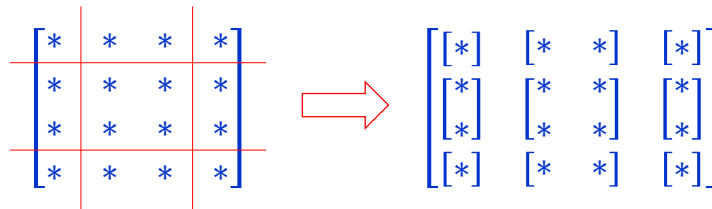
Here is the calculation:

$$\begin{aligned}
 & \begin{bmatrix} \begin{bmatrix} a_{11} & a_{12} \end{bmatrix} & \begin{bmatrix} a_{13} \end{bmatrix} \\ \begin{bmatrix} a_{21} & a_{22} \end{bmatrix} & \begin{bmatrix} a_{23} \end{bmatrix} \\ \begin{bmatrix} a_{31} & a_{32} \end{bmatrix} & \begin{bmatrix} a_{33} \end{bmatrix} \end{bmatrix} \begin{bmatrix} \begin{bmatrix} b_{11} \end{bmatrix} & \begin{bmatrix} b_{12} \end{bmatrix} \\ \begin{bmatrix} b_{21} \end{bmatrix} & \begin{bmatrix} b_{22} \end{bmatrix} \\ \begin{bmatrix} b_{31} \end{bmatrix} & \begin{bmatrix} b_{32} \end{bmatrix} \end{bmatrix} \\
 &= \begin{bmatrix} \begin{bmatrix} a_{11} & a_{12} \end{bmatrix} \begin{bmatrix} b_{11} \end{bmatrix} + \begin{bmatrix} a_{13} \end{bmatrix} \begin{bmatrix} b_{31} \end{bmatrix} & \begin{bmatrix} a_{11} & a_{12} \end{bmatrix} \begin{bmatrix} b_{12} \end{bmatrix} + \begin{bmatrix} a_{13} \end{bmatrix} \begin{bmatrix} b_{32} \end{bmatrix} \\ \begin{bmatrix} a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{21} \end{bmatrix} + \begin{bmatrix} a_{23} \end{bmatrix} \begin{bmatrix} b_{31} \end{bmatrix} & \begin{bmatrix} a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{22} \end{bmatrix} + \begin{bmatrix} a_{23} \end{bmatrix} \begin{bmatrix} b_{32} \end{bmatrix} \\ \begin{bmatrix} a_{31} & a_{32} \end{bmatrix} \begin{bmatrix} b_{21} \end{bmatrix} + \begin{bmatrix} a_{33} \end{bmatrix} \begin{bmatrix} b_{31} \end{bmatrix} & \begin{bmatrix} a_{31} & a_{32} \end{bmatrix} \begin{bmatrix} b_{22} \end{bmatrix} + \begin{bmatrix} a_{33} \end{bmatrix} \begin{bmatrix} b_{32} \end{bmatrix} \end{bmatrix} \\
 &= \begin{bmatrix} (a_{11}b_{11} + a_{12}b_{21}) + a_{13}b_{31} & (a_{11}b_{12} + a_{12}b_{22}) + a_{13}b_{32} \\ (a_{21}b_{11} + a_{22}b_{21}) + a_{23}b_{31} & (a_{21}b_{12} + a_{22}b_{22}) + a_{23}b_{32} \\ (a_{31}b_{11} + a_{32}b_{21}) + a_{33}b_{31} & (a_{31}b_{12} + a_{32}b_{22}) + a_{33}b_{32} \end{bmatrix}
 \end{aligned}$$

As expected.

Comments

You divide a matrix into blocks by grouping together the rows, and grouping together the columns. You can think of this as putting horizontal and vertical dividers across the matrix.



Writing linear systems as matrix equations

Video 5.1.7

Of course, the heart of linear algebra is the solving linear equations.

Does all this matrix algebra we have just developed bring anything new to this fundamental problem?

Obvious but important fact:

An n -tuple $(x_1, \dots, x_n) \in \mathbb{R}^n$ solves the following system of m linear equations in n variables $x_1, \dots, x_n$

$$\begin{array}{rcl} a_{11}x_1 + \cdots + a_{1n}x_n & = & b_1 \\ \vdots & & \vdots \\ a_{m1}x_1 + \cdots + a_{mn}x_n & = & b_m \end{array}$$



if and only if

The following matrix equation is true:

$$\begin{bmatrix} a_{11} & \cdots & \cdots & \cdots & a_{1n} \\ \vdots & & & & \vdots \\ a_{m1} & \cdots & \cdots & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ \vdots \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ \vdots \\ \vdots \\ b_m \end{bmatrix}$$

The a_{ij} and b_i are fixed.

What we need to show:

That the linear combination

$$\lambda(y_1, \dots, y_n) + (1 - \lambda)(z_1, \dots, z_n) = (\lambda y_1 + (1 - \lambda)z_1, \dots, \lambda y_n + (1 - \lambda)z_n)$$

is also a solution.

We will do this by checking the corresponding matrix equation:

$$A \begin{bmatrix} \lambda y_1 + (1 - \lambda)z_1 \\ \vdots \\ \lambda y_n + (1 - \lambda)z_n \end{bmatrix} \quad \leftarrow \text{We need to show this equals } \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

$$= A \left(\lambda \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} + (1 - \lambda) \begin{bmatrix} z_1 \\ \vdots \\ z_n \end{bmatrix} \right)$$

By the definitions of matrix addition and scalar multiplication of matrices.

$$= A \left(\lambda \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \right) + A \left((1 - \lambda) \begin{bmatrix} z_1 \\ \vdots \\ z_n \end{bmatrix} \right)$$

Because matrix multiplication is distributive.

$$= \lambda \left(A \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} \right) + (1 - \lambda) \left(A \begin{bmatrix} z_1 \\ \vdots \\ z_n \end{bmatrix} \right)$$

Because scalar multiplication of matrices respects matrix multiplication.

$$= \lambda \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix} + (1 - \lambda) \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

Because our assumption is that these matrix equations are true.

$$= (\lambda + (1 - \lambda)) \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

Because scalar multiplication is distributive.

$$= \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

Because

$$\lambda + (1 - \lambda) = 1$$

in $\mathbb{R}$.

So because the corresponding matrix equation is true, it follows that the given linear combination

$$\lambda(y_1, \dots, y_n) + (1 - \lambda)(z_1, \dots, z_n)$$

is also a solution. □

Corollary (This is a mathematical word for “consequence”).

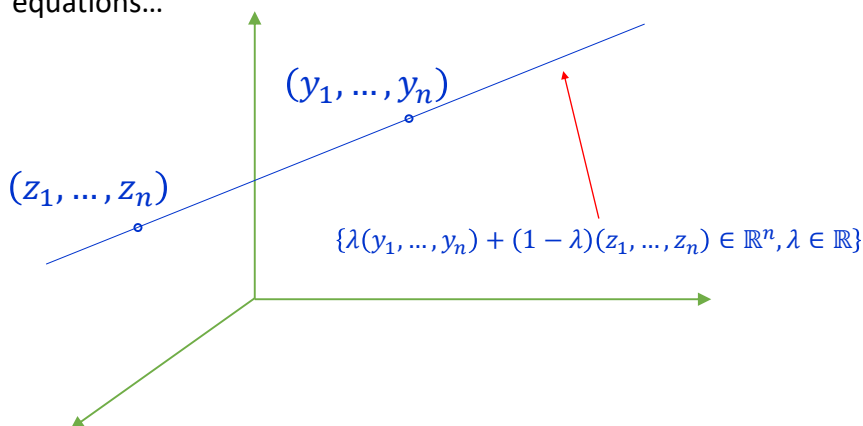
Every linear system has either:

- zero solutions.
- 1 solution.
- Infinitely many.

Because as soon as you have two different solutions, combining them as above produces infinitely many different solutions.

Illustration of the previous lemma.

It says that if two different points satisfy some system of linear equations...



... then every point on the whole line of points that goes between those two solutions must also solve the system.